clinanon: ICD-preserving anonymization of clinical encounters

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Summary. We remove private information from doctor–patient dialogues and clinical notes without touching what an ICD-10 coder needs. `clinanon` replaces only detected spans with typed tags. A clinical veto, built from ICD-10-CM, FDA and context rules, decides between the identifier and the clinical reading of ambiguous words. A 110M-parameter detector, distilled from a 27B open-weight LLM that ran only at development time, adds the cases that need context. We rebuilt the challenge's two metrics and computed them on four data sets. **Privacy:** all patient names and ages in the 207 challenge encounters are removed, and 94.4 % of the 1,493 known private values in held-out synthetic encounters. **Medical information:** ICD-10 coding F1 is unchanged (0.193 on the original, 0.195 on the anonymized text) and medical-concept recall is 99.7 %. Inference runs on a CPU, with no GPU.

1 Problem and design choices

The data are the 207 ACI-Bench encounters [1] with ICD-10-CM codes [2]. Each has a recorded dialogue, mostly lower-cased ASR output and sometimes without punctuation, and the physician's note. The organizers compare anonymized and original text on two things: *privacy*, the proportion of privacy-related information removed, and *preservation of medical information*, the F1 of ICD-10 code prediction and medical-concept recall. Inference must run on-prem with at most 4 GB of VRAM.

Two properties make the task hard. First, the same token can be an identifier or a diagnosis: *Mr. Parkinson* vs. *Parkinson's disease*, the occupation *farmer* vs. *farmer's lung* (J67.0), or the drug *Toprol* that NER tags as an organization. Second, names in the dialogue carry no capitalization: *"so , andrew , what's going on"*. We fixed three design choices early:

- **Minimal diff.** Only detected spans change. There is no generative rewriting, because every character outside a span is utility we cannot afford to lose. Speaker tags and section headers are protected.
- **Typed tags, not surrogates.** Realistic surrogates [3] and generalized values ("in his 50s") are themselves flagged by a detector-based scorer; `[AGE]` is not. The tag also tells a coder what kind of information was there.
- **An explicit taxonomy.** It has 12 categories, each grounded in HIPAA Safe Harbor [4], the i2b2 2014 guidelines [5], GDPR quasi-identifiers, TAB [6] or the Stanford label set [7]. One YAML file is both documentation and configuration. Some things are kept on purpose: relative time and durations, clock-face positions, types of place (external-cause codes need them), care-team roles and relatives, and occupations named in a diagnosis.

2 Method

2.1 Candidate detection

Rules are precise by construction. A date needs two calendar parts or a month word, so murmur grades (2/6) and pain scores (7/10) never match. A name needs a cue: a title, a greeting, a dictation header, or "X is a 59-year-old". The rules also cover formats ACI barely contains but other data will: international and spoken phone numbers, spoken e-mails, social-security and insurance numbers, street addresses and postcodes, hospitals and clinics, and labelled form fields ("MRN:", "SURGEON:"). A GeoNames gazetteer (90k names, filtered against dictionary and clinical words) matches only after a place cue: *"moved here from bergamo"*.

spaCy NER runs through Microsoft Presidio on a truecased copy of the dialogue: words the note capitalizes mid-sentence are capitalized in the dialogue too. The copy has the same length, so offsets map back 1:1. The **distilled student** is described in Sec. 2.3.

2.2 Clinical veto, propagation, explanations

Spans from unambiguous context, such as a title or a dictation header, are *strong*. Spans from NER, the student and the gazetteer are *weak*, and the veto discards a weak span when either of two things holds:

- its context is clinical: an eponym before a head noun (*disease*, *sign*, *test*), "positive Spurling's", or a dose right after a name, which marks a drug;
- all of its tokens are clinical or ordinary vocabulary (ICD-10-CM 2025 descriptions, FDA drug names, curated terms).

Confirmed names, facilities and places then propagate across both fields, including spoken short forms (*East Metropolitan Hospital → east metro*). A name that is also a clinical term only spreads where a title marks it, so "my neighbour mrs. thrombosis" does not erase *deep vein thrombosis*. Every decision, including each rejected candidate, is written with an explanation: evidence, confidence, the taxonomy category and its source.

2.3 Distillation from a 27B open-weight teacher

The VRAM limit applies to inference, not to development. Our teacher, Qwen3.8-27B [8], ran with vLLM [9] on two rented GPUs.

- **Labelling.** The teacher copies a chunk with inline XML tags. We accept a chunk only if removing the tags reproduces the input exactly, which makes the span alignment exact (1,274 of 1,291 chunks). It labelled the 140 non-eval encounters; the 67 eval encounters stay out of training.
- **Generation.** It wrote 1,066 encounters: 25 settings, varied patient backgrounds, 6 note formats, and deliberate traps (eponyms, drug brands, relative time). Tags written during generation marked *Parkinson's* and

				Privacy: removed		Medical information kept		
Data set	Docs	Source	Labels	known values	judge	ICD-10 F1 orig./anon.	concepts	own codes
ACI-Bench eval	67	real	names, ages (gold)	100 %	84.2 %	0.193 / 0.195	99.7 %	100.0 %
ACI-Bench other	140	real	names, ages (gold)	100 %	86.6 %	–	99.8 %	100.0 %
Synthetic test	40	ACI + inserted	1,493 values (gold)	**94.4 %**	–	–	99.3 %	100.0 %
Unseen-formats test	116	teacher-written	4,107 values (silver)	89.0 %	–	–	98.2 %	99.0 %

Table 1: The four evaluation data sets and both challenge metrics for the submitted configuration (rules, spaCy and the student). Concepts: ICD-10-CM and FDA vocabulary; own codes: words of the record's ICD-10 descriptions. "Judge" is the Presidio stand-in; it also counts eponyms and drugs as names, so it understates privacy. ICD-10 F1 is measured on the eval set only.

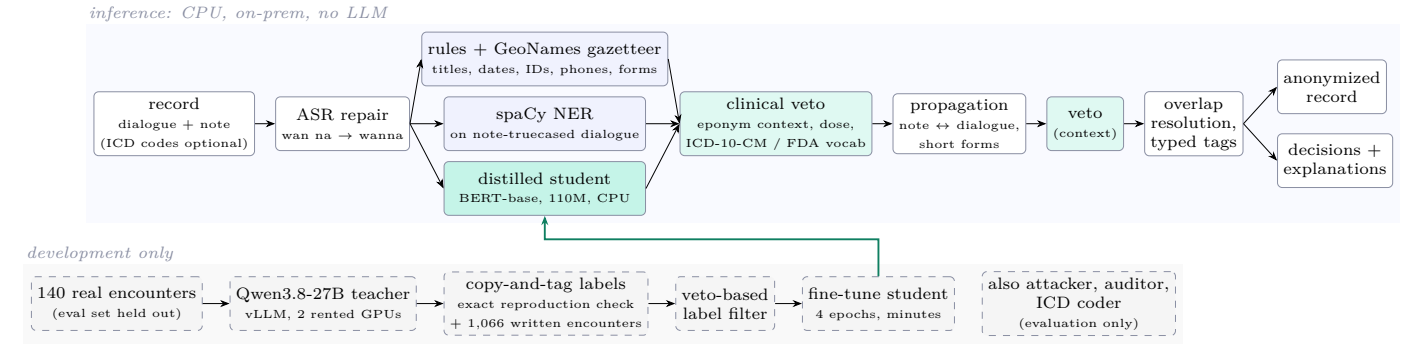


Figure 1: Architecture. Top: the shipped pipeline. Rules, spaCy and the distilled student propose spans; the veto keeps clinical readings; propagation spreads confirmed names; every decision is logged with an explanation. Bottom: the offline distillation that trains the student.

In: `Mr. Parkinson is a 67-year-old retired teacher with Parkinson's disease, seen today in Zürich on 14.05.2024. Continue Toprol 50 mg daily.`
Out: `Mr. [PATIENT] is a [AGE] retired [PROFESSION] with Parkinson's disease, seen today in [LOCATION] on [DATE]. Continue Toprol 50 mg daily.`
In/Out: `i moved here from bergamo last spring → i moved here from [LOCATION] [DATE]`
"Parkinson" is a person's name (direct identifier; HIPAA, I2B2, TAB); evidence: it is the subject of a sentence stating an age. Candidate rejected: "Parkinson" was proposed as name (evidence: the same name was identified elsewhere), but it is followed by a clinical head noun, so it names a clinical concept. Candidate rejected: "Toprol" was proposed as organization (spaCy), but it is followed by a dose, so it is a drug.

Figure 2: Real pipeline output on the ambiguity the task describes, with three of its explanations. "today" is kept on purpose (relative time).

ibuprofen, so the generated text is labelled in a separate pass. A filter derived from the veto removes eponyms, drugs and ICD vocabulary from the labels.

- **Student.** The Stanford de-identifier [7] (BERT-base, 110M, uncased, so ASR and notes look alike), fine-tuned for 4 epochs in minutes. It reaches span F1 0.974 against the teacher on held-back documents. This is targeted distillation of an LLM into a small NER model [10, 11].

Training setup. Teacher and student ran on one rented Vast.ai instance with two NVIDIA RTX PRO 6000 GPUs, the teacher served by vLLM. The student was fine-tuned on 1,161 documents (9,545 chunks of at most 512 tokens) for 4 epochs (learning rate $5 \cdot 10^{-5}$, effective batch 16, bf16) in about three minutes. Inference needs none of this: the student runs on a laptop CPU.

Reviewing its edits on real encounters changed how we use it. It had tagged *farm* inside *farmer*, and lab values as ages. It now contributes only names, places, organizations and occupations, with whole-word spans and a confidence of at least 0.75, and it stays behind the veto.

3 Evaluation data and metrics

We never had the organizers' pipeline, so we rebuilt both metrics. Each compares the anonymized text with the original, and we computed them on the four data sets of Table 1.

- **Privacy: proportion of private information removed.** A known private value counts as removed only if every letter and digit of it lies inside a tag ("+41 79 `[CONTACT]`" is a leak). ACI labels only the patient's names and age, so on real data we add a stand-in for the organizers' detector: Presidio runs on the original, and we count the share of its detections that no longer occur in the output.
- **ICD-10 F1.** The 27B teacher acts as a clinical coder on the original and on the anonymized eval set, and its codes are scored against the gold codes.
- **Medical-concept recall.** The share of medical-concept occurrences in the original that are still in the output. Concepts are the words of the ICD-10-CM 2025 descriptions and FDA drug names; we also report the words of each record's own ICD-10 code descriptions.

The two ACI-Bench sets are the real data. On the 67 *eval* encounters (the file we submit) we inspected misses, so they are not held out for the rules. The other 140 were used only to diff every rule change for regressions (this caught "um , my cousin" becoming a name that propagation spread to 50 places); the teacher's labels on them trained the student. The *synthetic test* gives each of 40 such encounters a new patient identity and 4–6 inserted administrative exchanges (birth date, phone, e-mail, address, insurer, employer, relatives, other doctors), worded by a local LLM with placeholders that Faker fills, so every value and its offsets are known. A disjoint dev split (40 encounters, 1,688 values) was used to look at misses; the test split was only scored. The *unseen-formats test* has 116 documents of types never used for training (radiology, pathology, ICU and operative reports, portal messages, triage calls), written and then labelled in a separate pass by the teacher; its labels are silver.

4 Results

Both metrics. Table 1 gives the result per data set. Names and ages are never the problem: real and inserted patient names are removed in 99–100 % of mentions. What remains are formats and context, mostly organizations and identifiers spoken in conversations. ICD-10 coding is unaffected: the coder's F1 is 0.193 on the original and 0.195 on our output. Concept recall is at least 98.2 % on every data set, and the words of each record's own codes are kept in full on real data.

Ablation. Table 2 and Fig. 3 switch off one component at a time.

- The *clinical veto* is the main lever between the two metrics. Without it the judge rises to 95.6 % (from 85.8 %), because Presidio counts eponyms and drugs as names, but 678 instead of 153 clinical words are lost (*Lisfranc fracture*, *Tylenol*). On known values it adds only 1.4 (synthetic) and 2.1 (unseen) points.
- *Rules* are the backbone (83.2 % alone, 47.3 % for spaCy alone); spaCy and the *student* add context. The student raises the synthetic test by 4.0 points (jobs 30→70 %, employers 72→84 %) for 11 more clinical words lost on 207 encounters. Zero-shot GLiNER [12] added 1.2 points but removed pronouns, anatomy (*l4 dermatome*) and

Variant	Privacy: removed synth.	unseen	judge	Concept recall ACI	lost
submitted (all three)	**94.4**	89.0	85.8	99.77	153
rules + student	93.8	88.1	84.4	99.77	154
rules + spaCy	90.4	86.1	85.7	99.79	142
rules only	83.2	77.3	81.0	99.81	125
spaCy only	47.3	44.3	53.8	99.96	29
rules + spaCy, no veto	92.0	88.4	95.5	99.02	652
submitted, no veto	95.8	91.1	95.6	98.98	678
rules + spaCy, no propagation	88.7	83.4	85.1	99.79	139
rules + spaCy, no truecasing	90.3	86.1	85.9	99.79	142
Presidio off the shelf*	66.4	71.2	(92.1)	96.88	2,076

Table 2: Ablation. Privacy on the synthetic and unseen-formats tests (known values) and with the judge on all 207 ACI encounters; concept recall and clinical words lost on the 207 ACI encounters. *Presidio is also the judge.

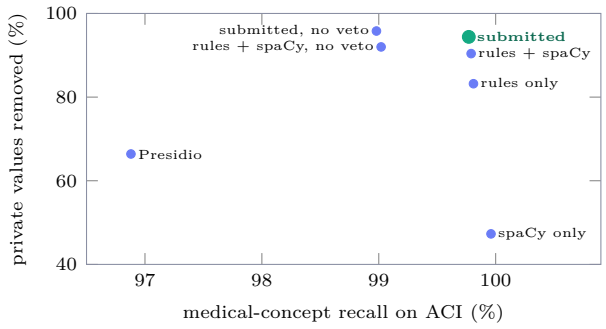


Figure 3: Privacy (synthetic test) against medical-concept recall (207 ACI encounters). Green: the submitted configuration. Removing the veto buys little true privacy for four times the clinical loss.

injury places (*the beach*), at 1.9 s per encounter.

- *Propagation* is what removes the last uncued patient names (gold names and ages 98.7 % without it). Truecasing no longer moves any metric by more than 0.2 points.
- Off-the-shelf Presidio removes fewer private values and loses 2,076 clinical words, 14 times as many as ours.

5 Bonus tasks

Adversarial attacks. The teacher acted as the attacker on our eval output (Table 3). In the third attack it first writes, from each original encounter, a profile of non-medical facts (job, hobbies, trips, events), then ranks all 67 anonymized encounters against each profile.

Attack on the final output	success	reference
recover removed values (776)	6.1 %	names 0.3 %
link via name, age, sex (1 of 10)	22.4 %	chance 10 %
link via non-medical profile (1 of 67)	73.6 %	originals 89.2 %

Table 3: Red-teaming with the 27B model. Recovery counts exact matches; ages are recovered within 3 years in 24 % of cases. Sex, kept on purpose, alone gives about 20 % in the 10-candidate test; chance for profile linking is 1.5 %.

Span-level anonymization removes identifiers, not the story, as the linkage and inference-attack literature predicts [13, 14]. Among 67 encounters almost every story is unique, and some details, such as how an injury happened, are exactly what external-cause codes need.

Explainability. 100 % of the 1,377 eval decisions carry an explanation, viewable in an HTML viewer. An LLM audit of a stratified sample of 120 decisions, weighted by category frequency, judged 93.6 % of removals and 100 % of kept spans correct, and about 91 % of the explanations accurate.

Efficiency. Table 4 lists four profiles on an Apple M4 Pro CPU. All use 0 GB of VRAM and remove 100 % of gold names and ages; parallel workers give identical output.

Profile	s/encounter	peak RAM	synthetic test
rules only	0.02	0.4 GB	83.2 %
rules + spaCy	0.25	1.4 GB	90.4 %
rules + student	0.39	2.1 GB	93.8 %
default (all three)	0.59	2.8 GB	**94.4** %

Table 4: Runtime, memory and privacy per profile.

Without ICD codes. The output is byte-identical on all 207 encounters. The record's codes only protect words that the global ICD-10-CM vocabulary already protects.

Robustness. The unseen document types are part of Table 1 (89.0 % on test, 89.9 % on dev). Labelled form fields ("Age: 75", "MRN:") were the main gap on dev.

6 What did not work

- **Reverse-engineering the scorer.** Stand-in judges disagreed widely on the same output (Presidio 84.2 %, an i2b2 de-identifier 97.1 % on the eval set), so we stopped imitating one and built the evaluation of Sec. 3 instead.
- **Zero-shot NER** had recall but no notion of the clinical boundary (Sec. 4).
- **The teacher's first labels** tagged eponyms and drugs; a separate labelling pass and the veto filter fixed this. Our first student read lab values as ages.

Limitations: the system covers English only, the robustness labels are silver, and narrative linkage remains.

7 Use of AI agents

Claude Code (Anthropic) was our coding assistant for implementation, experiments and documentation; the team made and reviewed every design and policy decision. The open-weight LLMs of Sec. 2.3 are used only at development time.

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